ChurnGuard: Protect customer revenue

Executive business report | Observed results and illustrative retention opportunities

THE PROBLEM

About 17 in every 100 customers churned.

Use churn predictions to focus retention effort where the risk is highest.

GOAL

Reduce avoidable customer loss without spending excessively on incentives.

Customer insights

The observed churn rate is 16.8%; customer segments and payment methods matter

About 336 in 2,000

At the observed 16.8% rate, around 336 out of 2,000 customers would have churned.

Customer segments

Super and Regular Plus together account for about 70% of customers.

Payment patterns

Debit and credit cards together account for about 72% of payment methods.

MEANING

Different customer groups may need different retention approaches.

How ChurnGuard works

A two-row workflow, with Streamlit alongside evaluation and strategy

1. Understand

Review records

2. Explore

Find patterns

3. Prepare

Clean data

4. Build

Train models

5. Evaluate

Test model

6. Segment

Group risk

7. Strategy

Costs and ROI

8. Streamlit

Score customers

Model Development -> Evaluation -> Segmentation -> Retention Planning -> Streamlit

Actual model evaluation

XGBoost performance on 2,252 held-out customer records

94% recall

Out of 100 customers who churned, the model found about 94.

86% precision

Out of 100 flagged as likely to churn, about 86 actually did.

96% accuracy

Across all customers, about 96 out of 100 predictions were correct.

OTHER MEASURED SCORES

F1-score 0.90 | PR-AUC 0.96 | ROC-AUC 0.99 | Cross-validation PR-AUC 0.9481.

Where is churn risk concentrated?

Actual test-set risk segmentation and estimated monthly revenue exposure

315 Critical accounts

Highest predicted churn risk; 68.36% of estimated revenue exposure.

57 High accounts

Next priority; 10.24% of estimated revenue exposure.

372 priority accounts

16.52% of test customers; together they hold 78.60% of estimated exposure.

OBSERVED PRIORITIZATION CHECK

Of the 20 highest-ranked test accounts, 19 actually churned. This is retrospective, not a live campaign.

Business impact: what could change?

Actual model outputs help the team decide whom to contact and how much to spend

Smaller target list

Focus on 372 of 2,252 test accounts rather than contacting everyone.

Exposure identified

2,525.38 in estimated probability-weighted monthly revenue at risk for priority accounts.

Better budget control

Compare the expected value of retaining customers against the cost of each offer.

IMPORTANT DISTINCTION

Prioritization is supported by test results. Prevented churn and financial savings have not been measured.

Retention offer costs

Illustrative campaign spending for 315 Critical and 57 High accounts | Dataset revenue units

Offer	Rate	Critical cost	High cost	Total cost
Personalized outreach	**5%**	**116.55**	**23.80**	**140.35**
Loyalty reward	10%	233.10	47.60	280.70
Discount voucher	15%	349.65	71.40	421.05
Cashback incentive	25%	582.75	119.00	701.75
Premium benefit	60%	1,398.60	285.60	1,684.20

Cost = total monthly revenue of each risk segment x offer rate. Currency is not specified in the dataset.

Top 5 illustrative ROI results: OPTIMISTIC

These are the notebook's top five segment-offer combinations, not Base Case outcomes

Segment / offer	Rate	Cost	Net benefit	ROI
Critical - Personalized outreach	**5%**	**116.55**	**3,441.46**	**2,952.78%**
High - Personalized outreach	5%	23.80	687.01	2,886.60%
Critical - Loyalty reward	10%	233.10	3,324.91	1,426.39%
High - Loyalty reward	10%	47.60	663.21	1,393.30%
Critical - Discount voucher	15%	349.65	3,208.36	917.59%

Optimistic success assumptions: Critical 45%, High 60%; 12 months retained revenue; 30% margin.

Decision: test offers before scaling

ROI is illustrative; the notebook includes Conservative, Base and Optimistic scenarios

Start with 5% outreach

Base Case ROI: 1,935% Critical; 1,891% High. Lowest-cost offer in the analysis.

Test 10-15% offers

Use loyalty rewards and vouchers selectively; compare against no-offer controls.

Avoid costly blanket offers

At 60% cost, Premium Benefit ROI is negative in the Conservative scenario.

WHAT TO MEASURE NEXT

Actual retention uplift, campaign cost, net benefit and realized ROI before wider rollout.

Base Case assumes 30% Critical and 40% High retention success; Conservative assumes 15% and 20%.